SAFE-CARE™ Healthcare AI Agent Safety Case Template

A structured, evidence-backed argument that an agent is safe enough for its intended use

SAFE-CARE Framework — authored by Yassen Eltayeb, Conefia · Version 1.1 · July 2026

Practical example: A safety case is the document you want in your hand when a hospital buyer asks: “How do you know this agent will not give unsafe advice?” It connects claims, risks, controls, evidence, and open issues.

Safety case structure

Section	What to write
1. Intended use	Who uses the agent, for what tasks, in what setting, with what limitations?
2. Safety claims	What must be true for the agent to be considered safe enough?
3. Hazard analysis	What can go wrong? Use the risk taxonomy.
4. Controls	Which guardrails, tool constraints, memory policies, and escalation rules mitigate each risk?
5. Evidence	Which tests, simulations, judge scores, clinician reviews, and production metrics support each claim?
6. Human oversight	When and how humans review, intervene, and approve releases.
7. Residual risk	What risks remain and why they are acceptable for intended use.
8. Monitoring plan	How production signals detect drift, incidents, and regressions.
9. Incident response	How severe failures are triaged, fixed, reported, and converted into regression tests.
10. Signoff	Product, AI, clinical, privacy/legal, and operations approval.

Safety claim examples

Claim	Evidence required
The agent does not provide prescription-like guidance.	Medication scenario suite, output guardrail logs, clinician review of failures.
The agent escalates red flags reliably.	100% pass on critical scenarios; production escalation monitoring.
The agent grounds medical claims in approved evidence.	RAG trace audit, citation judge score, clinician spot-check.
The agent writes sensitive health profile fields only after confirmation.	Tool-call logs, schema validation, confirmation UI screenshots.
The agent uses memory without stale or invented facts.	Time-series tests, contradiction scenarios, user correction logs.

Risk-control-evidence matrix

Risk	Control	Evidence	Owner
Missed red flag	Input classifier + output escalation check	Critical scenario suite pass rate	AI Safety
Unsafe medication advice	No-prescription prompt + output guardrail	HRT/medication scenario review	Clinical + AI
False citation	Citation support judge	Citation QA report	AI Evaluation
Bad profile write	Confirmation + schema validation	Tool trace audit	Backend + AI
Privacy leakage	Data minimization + retention policy	Privacy review + log audit	Privacy/Legal

Residual risk acceptance

Residual risk	Why it remains	Compensating control	Owner	Acceptance / expiry
Example: occasional false-positive	Sensitivity prioritized for a high-risk	Human review; user-friendly wording; monitor	Clinical safety lead	Accepted for pilot; review after 30 days

Residual risk	Why it remains	Compensating control	Owner	Acceptance / expiry
escalation	symptom class	escalation rate		
[Risk]	[Rationale]	[Control]	[Owner]	[Decision/date]

Safety claim pattern

Write every claim as: "For [intended use and population], configuration [version] is acceptably safe with respect to [specific hazard] because [controls] achieved [measured evidence], subject to [limitations and monitoring]." This prevents vague claims such as "the agent is safe."

Required review questions

- What evidence would falsify this safety claim?
- Which user group or context is not represented in the evidence?
- What happens when retrieval, tools, or escalation services are unavailable?
- Who can stop the system, and how quickly?
- When does the safety case expire or require re-approval?

References

[Case Study] Eltayeb, Y. (2026). Menopause-support companion design case study. In SAFE-CARE public release repository, v1.0. Conefia. GitHub/Zenodo. https://doi.org/10.5281/zenodo.21330842

[Conference Outline] Eltayeb, Y. (2026). Building Safe Conversational AI Agents for Healthcare: Conference talk outline. In SAFE-CARE public release repository, v1.0. Conefia. GitHub/Zenodo. https://doi.org/10.5281/zenodo.21330842

[WHO 2021] World Health Organization. Ethics and governance of artificial intelligence for health. 2021. https://www.who.int/publications/i/item/9789240029200

[WHO 2025] World Health Organization. Ethics and governance of artificial intelligence for health: guidance on large multi-modal models. 2025. https://www.who.int/publications/i/item/9789240084759

[NIST AI RMF] National Institute of Standards and Technology. AI Risk Management Framework. https://www.nist.gov/itl/ai-risk-management-framework

[NIST AI 600-1] National Institute of Standards and Technology. Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (NIST AI 600-1). U.S. Department of Commerce, July 2024. https://doi.org/10.6028/NIST.AI.600-1

[FDA GMLP] U.S. Food and Drug Administration. Good Machine Learning Practice for Medical Device Development: Guiding Principles. https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles

[FTC HBNR] Federal Trade Commission. Complying with FTC's Health Breach Notification Rule. https://www.ftc.gov/business-guidance/resources/complying-ftcs-health-breach-notification-rule-0

Public framework for educational and engineering use. It is not medical, legal, privacy, security, or regulatory advice; qualified review is required for each deployment.

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